Flipped Classroom Method in Higher Education: A Case of Kazakhstan

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Abstract. This study is set to explore flipped classroom method in Kazakhstan higher education. Teachers from eleven universities participated in the survey. Observation, questioning, counting, and comparison were used as the main research methods. The results are reported with respect to distribution of the flipped classroom method in the system of higher education in Kazakhstan, along with the influence of social factors on the use by teachers of the flipped classroom method, areas of knowledge, where the flipped classroom method is used for teaching in the higher education of Kazakhstan, the main reasons for using the flipped classroom method in the universities, the main reasons for not using the flipped classroom method in Kazakhstani universities, as well as description of difficulties in using the flipped classroom method. The results of our study can be beneficial for educators and researcher to guide their future studies on flipped classroom, especially their design and implementation.

Keywords: Flipped classroom · Kazakhstan · Higher education · Digital competence

1 Introduction

The relevance of the study of the application of the flipped classroom method in the system of higher education in our country is due to the current situation in the world. Due to the COVID-2019 spread, millions of students have interrupted traditional education. In this unstable environment, it is important that students continue the educational process. Around the world, scientists, methodologists, and information technology specialists are applying a creative approach to online options to ensure safety and health of students and provide them with the opportunity for quality distance learning. In this regard, the purpose of this work is to analyze the current state of the use of the flipped classroom method in Kazakhstani universities as one of the methods that can improve the quality of the educational process in distance and traditional education.

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As a result of the pandemic, education has changed dramatically – e-learning on digital platforms has markedly grown. The L.N. Gumilyov Eurasian National University accepted the challenge and, like most universities, switched to distance learning on March 16, 2020, which gave an impetus to a new round of digitalization of the university. The university tested the "Task" module in Microsoft Teams and is included in the current configuration of the Automation and Information System (AIS) "Platonus", and users have been active in the Learning Management System (LMS) "Moodle". The university departments used various subsystems and services such as Platonus, Moodle, Microsoft Teams, Hangout, Zoom, Webex Meeting, Webex Training and Google Classroom.

The flipped classroom method takes a new look at the traditional model: homework is done in class and lectures are watched at home. The purpose of flipped classroom is to provide students with a deeper learning experience as they study the material themselves, as recommended by their instructors. This is an educational concept in which the student acquires knowledge independently and in cooperation with the teacher. This approach allows managing the time, place, pace and path of learning the material. Among its features, the following aspects can be distinguished: individualization of learning, saving time in the classroom, variability of educational trajectories, asynchronous learning, new interaction opportunities, shortening the chain between the student and the learning content [1].

The new approach to learning began to receive support with further advances in technology. Two Colorado High School teachers Jonathan Bergmann and Aaron Sams discovered a software device in 2007 that allowed them to voice and record PowerPoint presentations.

Around the same time, financial analyst Salman Khan started making short math tutorials for his nephew, which he posted on YouTube.

Through these and other similar initiatives, the idea of providing learning materials to students through short online videos has begun to seep into higher education. Educators have also realized that by providing theory material via online video outside of the classroom, they can free up valuable time in the classroom for more productive and effective learning activities. This change in teaching practice became known as the flipped classroom. The main ideas of this technique were outlined by the authors in the book "Flip Your Classroom: Reach Every Student in Every Class Every Day" [2]. It was published in the original language in 2012 in the USA.

Since that time, there has been a stormy interest in the flipped classroom method in the world. There are various scientific studies [3–6]. These papers describe the core idea of the flipped classroom: students explore online resources outside of class so that classroom time can be devoted to developing practical skills on a particular topic. Studies have shown that this method leads to improved learning outcomes [7].

An analysis of studies related to the study of the flipped classroom method shows that they are conditionally combined into three large groups: 1) works related to the use of the flipped classroom method in schools (the largest group), 2) a flipped classroom in a university (the smallest group), 3) the use of this method in various fields of knowledge. And it should be noted that the issue of using the flipped classroom method is very closely related to the possibilities of using new technologies in general in the learning process. Thus, Greenfield suggests that since students are now constantly working with

technologies that are becoming more and more visual, it is not entirely correct for teachers to deliver and evaluate knowledge using outdated strategies [8]. For some scholars, technology has not only changed the way students learn, but has actually redefined the purpose and meaning of education itself. Thus, Vlieghe writes that the learning process must be understood in tune with Kant's idea – only through education do we become who we are. This means that the use of different educational technologies may be accompanied by the emergence of a completely different type of human subjectivity [9]. In essence, the goal of J. Bergmann and A. Sams was to use technology to maximize the effectiveness and efficiency of study time.

In the modern "piggy bank" of methodological recommendations there are a lot of works describing advantages of the flipped classroom method. The popularity of this teaching method is increasingly growing, and a variety of opinions and observations are expressed regarding its effectiveness [10]. One of its benefits is increased time efficiency. Since in a flipped classroom, video versions of materials are usually viewed as homework, reading articles on the topic of the upcoming lesson, students will be ready for active study of the material [11]. It also allows teachers to devote more time to individual work with students, which creates additional opportunities to test understanding and assimilation of the material being studied. Another time-saving benefit is that once teachers have recorded and documented course materials online, or provided links to related resources, students can easily refer to them in the future, saving them time to review.

Many researchers believe that the greater involvement of students in the learning process may be due to the fact that in an inverted classroom, the material is presented through digital media, which are easier for students to familiarize with [8]. Considering that most of today's youth grew up relying on web-based forms of information consumption and communication, it is possible that computer-based learning methods find a more natural response for them. Another explanation for the increase in students' motivation to learn is seen by some scientists in the fact that video lectures are more interesting than traditional ones. This is based on research into human nature and physiology, which has shown that students typically lose interest in activities after about ten minutes. And the structure of the flipped classroom method involves the use of short video lectures to keep students interested [12].

If we turn to the analysis of domestic Kazakhstani studies on the flipped classroom method in the educational process, we can see progress from theoretical reviews to practical recommendations. So, between 2016 and 2019, many works were of a descriptive character. For example, the «NMC Horizon Report: 2015 Higher Education Edition" of the L. N. Gumilyov Eurasian National University briefly describes the experience of foreign universities and offers resources for introducing the flipped classroom into their own experience. The history of the appearance of this method abroad is also described in a number of other works [13]. In some studies, it is mentioned among new learning technologies [14]. In the collection "Trends of Modern Education", one of the chapters offers links to Internet resources that can be used with the flipped classroom method [15]. At the present stage, several articles can be singled out that describe the experience of using the flipped classroom in higher education institutions, and propose an algorithm for conducting classes [16], however, there are few such works. A review of studies of

Kazakhstani experience in the application of the flipped classroom method at a university shows the need for a systematic study of the application of this method, the formation of detailed practical recommendations in this area [17].

2 Research Methodology

The present work was organized in several stages. At the stage of studying the history of the issue, for a complete analysis of the existing references, methods of a theoretical level were used – reviewing and generalization, content analysis. The research program was developed on the basis of a preliminary study of the material on the use of the flipped classroom method in Kazakhstan and abroad. As a result of studying the existing experience, the basic concepts of research were determined – the flipped classroom method and its main steps. As part of the content analysis of theoretical sources and personal observations, the goal was determined – to identify the level of use of the flipped classroom method in the universities of Kazakhstan.

To collect data for the study, empirical level methods were used – observation, questioning, counting, and comparison. At this stage, an individual correspondence survey was organized using a Google form [18].

The results obtained in the process of content analysis were supplemented and concretized with the help of a sociological survey. The survey was organized using a computer questionnaire, which was developed and posted by us on Google forms [18]. It was distributed among the universities of Kazakhstan. Its purpose was to collect data on the use of the flipped classroom method in Kazakhstani universities. To achieve it, grid-in questions, questions with multiple choice, dichotomous questions were formulated.

3 Results

3.1 Social Characteristics of Survey Participants

To learn the social status of the respondents, we proposed questions aimed at determining their age, gender, length of service, place of work and field of activity. In the general part of the questionnaire, we collected information about the age and gender characteristics of the participants.

Gender diversity is represented by 86.5% females and 13.5% males. This imbalance reflects an objective picture of the teaching staff (TS) of universities in Kazakhstan and, perhaps, this indirectly affects the use of the flipped classroom method in the higher education system [19], women often have less access to information and communication technology (ICT) than men. It is well known that when using the flipped classroom method, at least basic ICT skills are required. Perhaps the predominance of female contingent in the field of education also affects the level of dissemination of this method.

The age of the respondents ranged from 21 to 65 years.

In this part of our sociological survey, we also obtained the result for the place of work of our respondents. Based on these data, we cannot draw conclusions on the indicators of the use of the flipped classroom method in specific universities – only teachers from 11 universities responded to the survey. And the presence of a subjective factor is possible

here – respondents from the L. N. Gumilyov ENU are several times more than from other universities, since the authors of the article belong to this university and had the opportunity to involve more participants in the survey. Definitely, with such a number of respondents, our study does not claim to be a comprehensive study. However, the fact that 74 people from different universities showed interest in using this method in university education suggests that in our country there is a tendency to master this method to teach students to work independently in order to expand their knowledge.

By field of activity, survey participants were distributed as follows: foreign language teachers – 78.4%, humanities teachers – 14.9, natural sciences teachers – 2.7%, computer science and technology teachers – 2.7%, veterinary sciences teachers – 1.4%.

As can be seen from the distribution of percentages in this issue among the participants in the process, the degree of awareness is higher among foreign language teachers. There may be several reasons for this: (1) the first language in which the experience of using the flipped classroom began to spread was English; and (2) in the world practice there is a lot of literature on the use of this method in teaching a foreign language. Despite the fact that the authors and organizers of the sociological survey are English language experts, the questionnaire was distributed among teachers of various specialties, this is evidenced by the presence of 21.7% of the answers of the survey participants (humanities, natural sciences, computer science and technology, veterinary sciences). An analysis of the data on this issue allows us to draw attention to the expediency of a wide coverage of the advantages of the flipped classroom method in the learning process in all specialties of universities.

The majority of respondents have more than 10 years of experience in the field of education – 71.6%, there are also those whose experience is 25 years (1.3%). Less than 5 years – 12.2%, 5–10 years – 4.9%. The average work experience in the educational system was 9.5 years.

What does this indicator show for our study? Firstly, that experienced teachers participated in it, which allows us to talk about their familiarity with many teaching methods. Secondly, proficiency in ICT is required to apply the flipped classroom method. According to research, "it is the high ICT competence of a teacher that makes it possible to achieve high results in learning even in the absence of material and technical opportunities for acquiring the latest equipment and software" [20]. The experience of teachers for more than 10 years (71.6) seems to be favorable ground for mastering/developing digital competence, which, in turn, will allow the teacher to apply the flipped classroom method without any special obstacles, subject to the desire.

3.2 Understanding Flipped Classroom Method

One of the objectives of the study was to identify participants' perceptions of the flipped classroom method. To this end, participants were asked to choose a definition of the method. As shown by the statistical data obtained as a result of the survey, among the teaching staff of the Kazakhstani universities, the correct concept of the essence of the method was formed – 73% (54 respondents) indicated the correct definition of the flipped classroom. However, there is still work in this direction: if 27% of the survey participants also understand the idea of this method, they may begin to actively apply it. That is, the data on this result give grounds not only to state the understanding of this method by the

majority of university professors, but also to determine the education strategy for those who have not yet come across it even in theory.

3.3 The Use of the Flipped Classroom Method in Practice in Kazakhstani Universities

Next, we determine the level of use of our method in the universities in Kazakhstan. To achieve this goal, we have formulated 3 questions: 1. Have you ever used the flipped classroom method? 2. You are using the flipped classroom method because (please list all acceptable options) and 3. You are not using the flipped classroom method because (please list all acceptable options).

Data on the use of the flipped classroom method indicate that the vast majority of respondents (67.6%) use this method in practice. Less than twice (32.4%) do not use this method in the educational process. This suggests that the teaching staff of universities began to create the practice of developing skills of students' independent work. Traditional material supply – sage on the stage – when the teacher gives a lecture, and the students are passive, began to transform into an active way of gaining knowledge – guide on the side, that is, the teacher is not the ultimate truth, his/her role is transformed into tactful guidance of students' activities, into an assistant in acquiring new skills.

Reasons for Using the Flipped Classroom Method in Universities. When answering this question, the participants had the opportunity to select several answers.

Among the reasons that prompted the use of this method, in the first place was the release of classroom time for working on practical skills – 44.6% of respondents mentioned it. This positive reason is also highlighted in other flipped classroom studies. So, in a special report of the international organization Faculty Focus "Flipped Classroom Trends: A Survey of College Faculty, 2015", organized by the Community for the Empowerment of Higher Education (Faculty Focus, Madison, USA) 69.65% of college teachers, as a convincing reason for choosing this method, indicated the opportunity to move away from traditional lectures and pay attention to developing skills necessary for the student [21].

The second place was taken by an advantage as the fact that students receive new knowledge at their own pace, autonomously – 35.1%. In many works of domestic and foreign researchers, this undeniable advantage of this method has been repeatedly pointed out. The property of a flipped classroom to enable students to work in their own mode was noted by the majority of participants in our survey.

The third place among the reasons that prompted our respondents to turn to the method under study was occupied by the formation of critical thinking among students – 33.8%. It is quite natural that when students independently analyze the educational material, critical thinking develops well.

These 3 results give reason to hope that in the near future the vast majority of university professors in our country will teach students to acquire theoretical knowledge on their own, and classroom time will be devoted to acquiring practical skills by students.

The Main Reasons for Using the Flipped Classroom Method. In the first place, the respondents identified such an advantage of this method as "more time to develop practical skills in class time" – 44.6%. The answer "students become creators of knowledge" has the least percentage.

The reasons why some survey participants do not use the method under study are formulated on the basis of a content analysis of theoretical sources, observations and personal practice. Table 1 contains all the answers to the question about the reasons for not using the flipped classroom method in practice. A large percentage of respondents were those who are not convinced of the independence of our students – 36.5%.

Table 1. Reasons for not using the flipped classroom method

Reasons for not using the method	%
Not all students are independent enough	36.5
I don't know about this method	13.5
My course cannot be taught with this method	6.8
I don't believe in its effectiveness	5.4
It weakens the teacher's role	5.4
Ignorance of new technologies hinders	4.1
The amount of work will increase	5.1
Other (I use, I don't use)	24.2

3.4 Difficulties in Using the Flipped Classroom Method

To study the problems of introducing this method into the system of higher education, we identified such an item in a sociological survey as the identification of problematic stages in the application of the flipped classroom. Respondents were given a list of possible difficulties in using this method. The biggest difficulty is the motivation of students to study the material at home on their own – this was indicated by 33 people (44.6%); the development of assignments that expand the possibilities for students' further self-study turned out to be difficult for 16 people (21.6%); development of tasks and resources – for 12 people (16.2%); determination of the volume of educational material, goals and strategies – for 10 people (13.5%); current and final assessment of students' knowledge – for 9 people (12.2%); development of assignments for classroom work – for 6 people (8.1%); and 6 people (8.1%) do not experience any difficulties.

Also, one of the reasons why this method is not widely used is the fact that many teachers have an education received in a classical university, where there were no specialized courses aimed at mastering innovative teaching methods.

4 Discussion

In the process of studying the issue of using the flipped classroom method in the system of higher education in Kazakhstan, the following results were obtained.

Distribution of the Flipped Classroom Method in the System of Higher Education in Kazakhstan. Despite the number of participants in the survey (76 people), the

sociological survey showed in miniature the state of use of this teaching method in the university system of our country – an expansion of the trend towards the development and active use of the flipped classroom method to teach students to work independently to expand their knowledge.

Influence of Social Factors on the Use by Teachers of the flipped Classroom Method. The age of teachers does not create obstacles for mastering this method in practice. However, among the teaching staff of universities in Kazakhstan, the majority is made up of a female contingent, which, as you know, is not as active in mastering ICT as men (such a disproportion is observed all over the world). Therefore, this moment is an objective reason for the insufficiently high level of use of the flipped classroom method.

As the survey showed, the most favorable experience for the use of the flipped classroom technology for teachers is more than 10 years of experience. The effective application of this method in teaching requires not only successful mastery of ICT, but also methodological competence.

Areas of Knowledge, Where the Flipped Classroom Method is Used for Teaching in the Higher Education of Our Country. The degree of application of the flipped classroom method in the university system of our country is higher in the field of teaching the humanities, among which teaching foreign languages stands out. An analysis of the data on this issue allows us to draw attention to the expediency of a wide coverage of the advantages of the flipped classroom method in the learning process for almost all courses of universities. One of the goals of this method is to teach students to work independently with sources and to critically evaluate them.

The Main Reasons for Using the Flipped Classroom Method in the Universities of Our Country have been Identified. Among the main reasons that encourage teachers to use this method in teaching is the release of classroom time for developing practical skills. Such motivation for the application of this method, of course, increases the functional literacy of the acquired knowledge. However, another important criterion in education, which, in our opinion, took an undeserved third place is the development of students' critical thinking skills when using sources independently. This situation allows us to pay special attention to the emphasis that needs to be placed on the development of critical thinking when using the flipped classroom method.

The Main Reasons for Not Using the Flipped Classroom Method in Kazakhstani Universities have been Identified. One of the main reasons for rejecting this method is the belief of many teachers in the lack of independence of our students. This point of view is often based on the experience of traditional education that the teaching staff of Kazakhstani universities has. One of the ways to be convinced of the opposite, in our opinion, is the exchange of experience with teachers who actively and effectively use the flipped classroom method, demonstrating the ability of students to work independently. The teacher who decides to use the flipped classroom method must be ready for a lot of work, especially in the beginning.

Description of Difficulties in Using the Flipped Classroom Method. The stage of motivating students to study the material at home on their own turned out to be the most

difficult when using this method in the universities of our country. This is quite natural, since motivation is an area of a psychological order. And for many of our students, the traditional model of acquiring knowledge is still strong – the lecturer presents the material, the student listens to him/her. This creates certain difficulties at the initial stage. In our opinion, students should be immediately explained the principles of the flipped classroom method, and then given detailed instructions on each topic. T In the process of practical work, students gradually begin to show interest, since before them is no longer a bare theory, but its concrete application. This is how motivation is born.

5 Conclusions

Thus, by means of content analysis of sources, we briefly described the history of the emergence and application of the flipped classroom method in Kazakhstan and abroad. By means of a sociological survey, a certain research base was collected on the issue of using the flipped classroom method in higher education in Kazakhstan, which was analyzed.

Participation in the survey of people of different gender and age structure, different place of residence, direction of the sphere of professional interests, made it possible to ensure a certain representativeness of the survey. These studies in miniature showed the features of using the flipped classroom method in the system of higher education in Kazakhstan.

It should be noted that the use of this method has become more and more widespread in the universities of our country and, as sociological survey data have shown, arouses genuine interest in the category of teaching staff that works according to the traditional teaching model.

Also, one of the results of this work was the observation of certain gaps in the idea of the flipped classroom method among some of the teachers, which gives reason to suggest developing a strategy for teaching this method to those who want to use it in the learning process of students.

The results of our study indicate the need for a full comprehensive study of this issue in order to identify the strengths and weaknesses of using the flipped classroom method in the universities of Kazakhstan in order to organize further methodological assistance.

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